

Sri Lanka Institute of Information Technology



IE2044 – System Modeling and Prototyping

Design and Implementation of a 2-Digit Digital Thermometer Using LM35 and ICL7107

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Submission Date: 1st May 2026

TABLE OF CONTENTS

1. INTRODUCTION	3
1.1 Background	3
1.2 Objective	3
2. DESIGN METHODOLOGY	3
2.1 System Block Diagram	3
2.2 Working Principle	3
2.3 ICL7107 ADC Operation	4
2.4 Reference Voltage Calculation	5
2.5 Oscillator Design Calculation	6
2.6 Integrator Components	7
2.7 Dual Power Supply Design	7
2.8 PCB Design Methodology	7
3. COMPONENT SELECTION	9
3.1 LM35 (Through Hole)	9
3.2 ICL7107 (DIP-40)	9
3.3 Resistors (0805 SMD)	9
3.4 Capacitors	9
3.5 7-Segment Displays	9
4. PCB DESIGN CONSIDERATIONS	10
4.1 Analog Grounding (Star Ground)	10
4.2 Track Width Calculation	10
4.3 Decoupling Capacitors	10
5. RESULTS & TESTING	12
5.1 Proteus Simulation Results	12
5.2 Calibration Procedure	13
5.3 PCB Verification	13
5.4 Practical Hardware Implementation	13
5.5 Observed Issues During Hardware Testing	14
5.6 Technical Analysis of Instability	15
5.6.1 Breadboard Parasitic Effects	15
5.6.2 Ground Noise Problem	15
5.6.3 Power Supply Ripple	16
5.6.4 Long Signal Wires	16
5.7 Simulation vs Hardware Comparison	16
5.8 PCB Fabrication Status	17
6. DISCUSSION	17
6.1 Challenge 1: Unstable Display Readings	17
6.2 Challenge 2: Dual Power Supply Noise	18
6.3 Challenge 3: Reference Calibration Difficulty	19
6.4 Challenge 4: Breadboard Parasitic Effects	20
6.5 Challenge 5: DIP-40 Wiring Complexity	20
6.6 Challenge 6: Ground Loop Noise	21
6.7 Challenge 7: Fabrication Constraints	21
6.8 Lessons Learned	22
6.9 Improvements for Future Work	22
7. CONCLUSION	23
8. REFERENCES	23

1. INTRODUCTION

1.1 Background

Temperature measurement is essential in industrial, medical, and environmental applications. Digital thermometers provide higher accuracy, readability, and noise immunity compared to analog systems.

1.2 Objective

The objective of this project is to:

- Design a 2-digit digital thermometer
- Measure temperature range: 0°C to 99°C
- Use:
 - LM35 precision temperature sensor
 - ICL7107 A/D converter and 7-segment driver
- Display output on two 7-segment displays
- Implement complete system on PCB

2. DESIGN METHODOLOGY

2.1 System Block Diagram

Temperature → LM35 → Voltage Scaling → ICL7107 → 7-Segment Displays

↓

Dual ±5V Supply

2.2 Working Principle

Step 1: Temperature Sensing (LM35)

The LM35 outputs voltage proportional to temperature:

$$V_{OUT} = 10mV \times T(^{\circ}C)$$

Examples:

Temperature	Output Voltage
0°C	0 mV
25°C	250 mV
50°C	500 mV
99°C	990 mV

2.3 ICL7107 ADC Operation

The ICL7107 is a 3½-digit dual-slope integrating analog-to-digital converter with an internal 7-segment display driver. It directly converts an analog voltage into a corresponding digital display output without requiring a microcontroller.

Operating Principle

The ICL7107 operates in three main phases:

1. Auto-zero phase – Internal offset voltages are stored and compensated.
2. Signal integration phase – The input voltage (V_{IN}) is integrated over a fixed time period.
3. Reference de-integration phase – The integrator is discharged using the reference voltage (V_{REF}), and the time required is measured.

Because it uses dual-slope integration, the converter provides:

- High noise rejection
- Good accuracy
- Excellent stability against power supply variations

Relationship Between Input and Display

The displayed reading is proportional to the ratio of the input voltage to the reference voltage:

$$\text{Display} \propto \frac{V_{IN}}{V_{REF}}$$

Thus, the accuracy of the displayed temperature directly depends on:

- Stability of the input signal
- Accuracy of the reference voltage

Scaling for Temperature Measurement

Since the LM35 produces:

$$V_{OUT} = 10mV \times T(^{\circ}C)$$

To ensure:

$$10mV \rightarrow 1 \text{ display count}$$

The reference voltage was selected as:

$$V_{REF} = 100mV$$

With this configuration:

$$\frac{10mV}{100mV} = 0.1$$

which corresponds to one count per degree Celsius after appropriate display configuration.

This scaling ensures that the display directly shows the temperature in °C without additional signal amplification.

2.4 CRITICAL CALCULATION – REFERENCE VOLTAGE

Given:

LM35 output = 10mV per °C

We want:

$$10mV = \text{Display increment of } 1$$

Using formula:

$$\text{Display} = \frac{V_{IN}}{V_{REF}} \times 100$$

To make:

$$10mV \rightarrow \text{Display} = 1$$

So:

$$1 = \frac{0.01}{V_{REF}} \times 100$$

$$V_{REF} = 0.1V = 100mV$$

Therefore:

$$\boxed{V_{REF} = 100mV}$$

This is why the potentiometer is adjusted to exactly 100mV.

2.5 Oscillator Design Calculation

The ICL7107 requires an external RC network to generate its internal clock frequency. This clock determines the integration timing and overall conversion rate of the ADC.

Selected Component Values

From the schematic:

$$R = 100k\Omega$$

$$C = 100nF$$

Oscillator Frequency Formula

According to the ICL7107 datasheet, the oscillator frequency is approximately:

$$f = \frac{1}{2.2RC}$$

Frequency Calculation

$$f = \frac{1}{2.2 \times 100k\Omega \times 100nF}$$

Substituting values:

$$\begin{aligned} f &= \frac{1}{2.2 \times 100000 \times 100 \times 10^{-9}} \\ f &= \frac{1}{2.2 \times 0.01} \\ f &= \frac{1}{0.022} \\ f &\approx 45.45Hz \end{aligned}$$

Significance of Oscillator Frequency

This frequency determines:

- Integration period
- Conversion speed
- Noise rejection characteristics

A frequency around 45 Hz ensures stable operation and adequate rejection of low-frequency noise components.

2.6 Integrator Components

We used:

- $R = 100\text{k}\Omega$
- $C = 0.47\mu\text{F}$

These set the integration time and noise rejection (especially 50Hz power noise rejection).

Include explanation:

Dual slope integration cancels AC noise, improving stability.

2.7 Dual Power Supply Design

ICL7107 requires:

$+5V$ and $-5V$

You used:

ICL7660 Voltage Converter

It converts:

$+5V \rightarrow -5V$

Using charge pump capacitors:

- $10\mu\text{F}$
- $10\mu\text{F}$

Working principle:

It switches capacitors to invert voltage polarity.

Include equation:

$$V_{OUT} \approx -V_{IN}$$

2.8 PCB Design Methodology

After successful simulation, PCB design was performed in EasyEDA following these steps:

1. Import schematic to PCB layout
2. Arrange components logically:
 - Sensor section separated from display section
 - Power section isolated

3. Implement star grounding:
 - LM35 ground and ADC analog ground connected at single point
4. Minimize analog trace length
5. Align 7-segment displays horizontally
6. Perform Design Rule Check (DRC)

The final PCB layout is shown in Figure 3, and 3D visualization in Figure 4.

EasyEDA Schematic Diagram

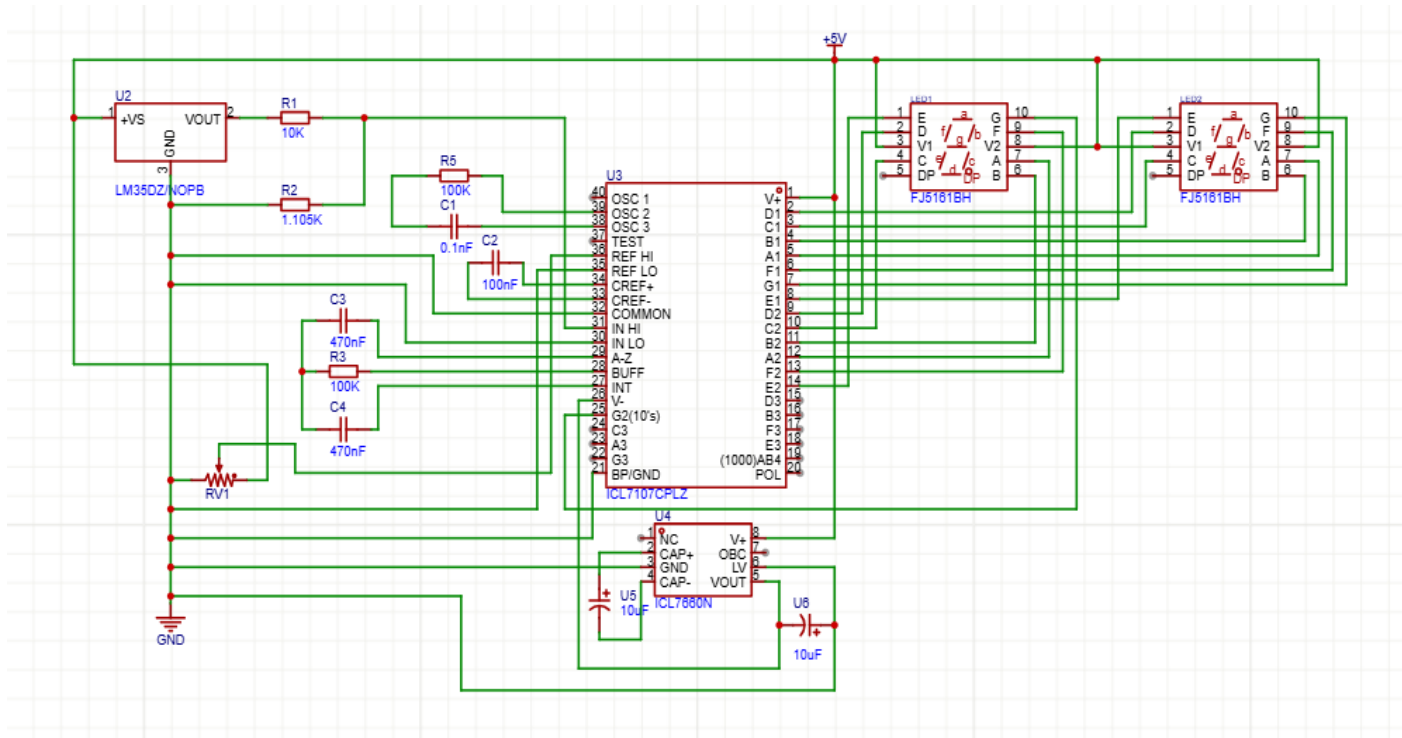


Figure 2: Complete schematic diagram of the digital thermometer designed in EasyEDA showing LM35 sensor, ICL7107 ADC, dual power supply, and 7-segment displays.

- The complete circuit schematic implemented in EasyEDA is shown in Figure 2. The design includes the LM35 temperature sensor, reference voltage adjustment circuit, oscillator components, dual supply generation, and display connections.

3. COMPONENT SELECTION

3.1 LM35 (Through Hole)

Reasons:

- Linear output
- No calibration required
- 10mV/°C sensitivity
- $\pm 0.5^\circ\text{C}$ accuracy
- Easy PCB mounting

3.2 ICL7107 (DIP-40)

Reasons:

- Direct 7-segment drive
- No microcontroller needed
- Integrated ADC + display driver
- Simplifies design

3.3 Resistors (0805 SMD)

- Compact
- Professional PCB appearance
- Good thermal stability

3.4 Capacitors

- 100nF for decoupling
- 0.47 μF for integration
- 10 μF for charge pump

3.5 7-Segment Displays

- Common Anode type
- Compatible with ICL7107
- Horizontally aligned (as required)

4. PCB DESIGN CONSIDERATIONS

4.1 Analog Grounding (Star Ground)

You implemented:

Single connection point between:

- LM35 ground
- ICL7107 analog ground

Why?

To prevent:

- Display switching noise
- Ground loop errors
- Reading jitter

Explain:

$$V_{error} = I_{noise} \times R_{ground}$$

Star grounding minimizes this error.

4.2 Track Width Calculation

If display current $\approx 15\text{mA}$ per segment

Worst case (all segments ON):

$$I_{max} \approx 8 \times 15\text{mA} = 120\text{mA}$$

Using PCB calculator:

For 120mA, 1oz copper:

Minimum width $\approx 0.3\text{mm}$

✓ Your tracks are wider $\rightarrow$ Safe design

4.3 Decoupling Capacitors

100nF placed near IC power pins to reduce:

$$V_{noise} = L \frac{di}{dt}$$

Prevents oscillation.

PCB Layout (2D Routing View)

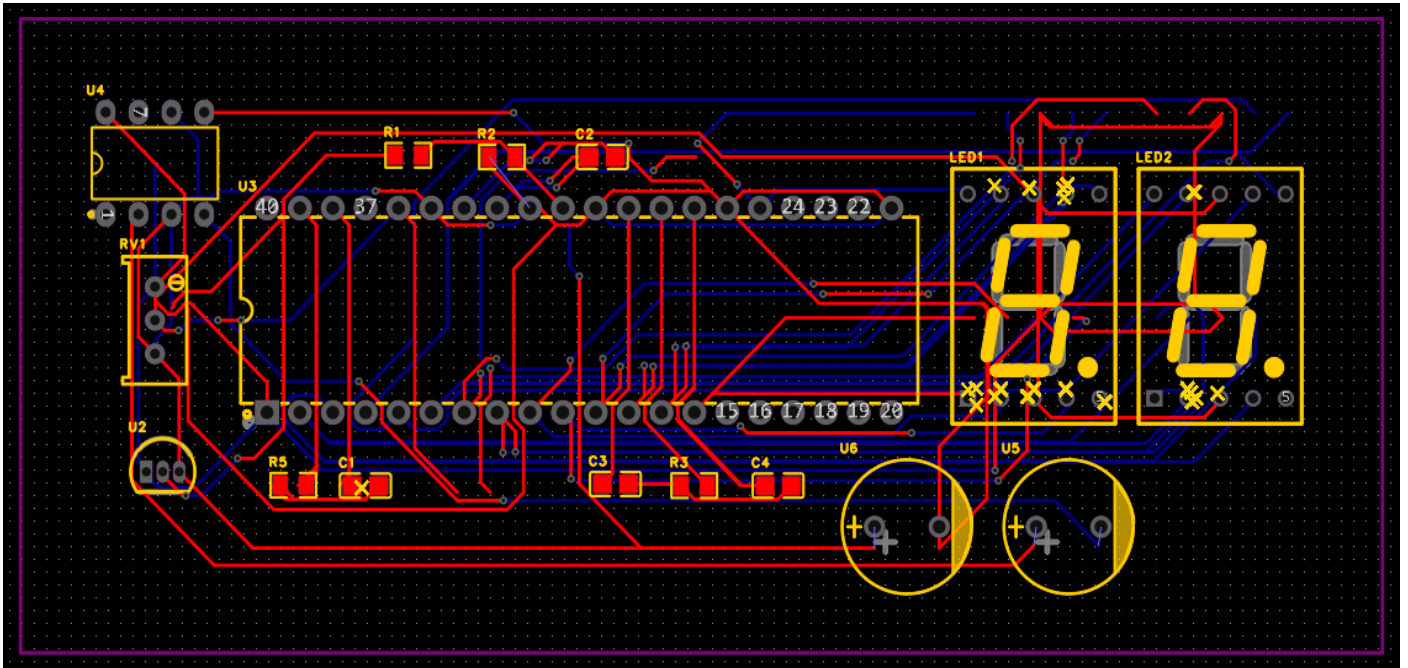


Figure 3: PCB layout (top view) showing component placement, analog routing, and star grounding implementation.

- The final PCB layout generated in EasyEDA is shown in Figure 3. Special attention was given to analog grounding, short signal traces, and proper alignment of the 7-segment displays.

3D PCB Design View

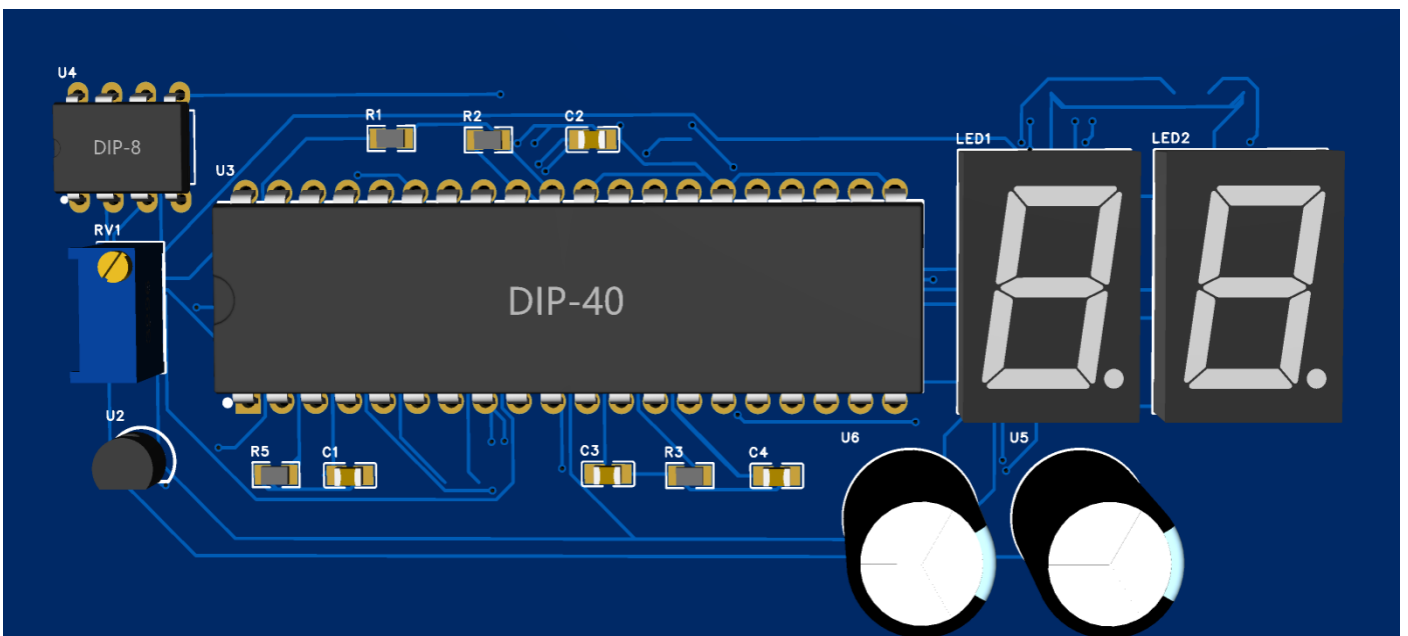


Figure 4: 3D visualization of the designed PCB illustrating component placement and horizontal alignment of the two 7-segment displays.

5.2 Calibration Procedure

1. Set LM35 to known temperature (e.g., 25°C)
2. Adjust potentiometer
3. Measure $V_{REF} = 100\text{mV}$
4. Confirm display matches temperature

5.3 PCB Verification

- No DRC errors
- Proper alignment of displays
- Clean routing
- Compact design

5.4 Practical Hardware Implementation (Breadboard Testing)

Although the circuit was fully verified in Proteus simulation and designed in EasyEDA, the practical implementation was carried out on a breadboard for hardware testing.

During hardware testing, the system powered up successfully; however, the displayed temperature values were:

- Not stable
- Fluctuating continuously
- Sensitive to touch and nearby electrical interference

The output did not consistently match the expected temperature value.

Breadboard Hardware Implementation

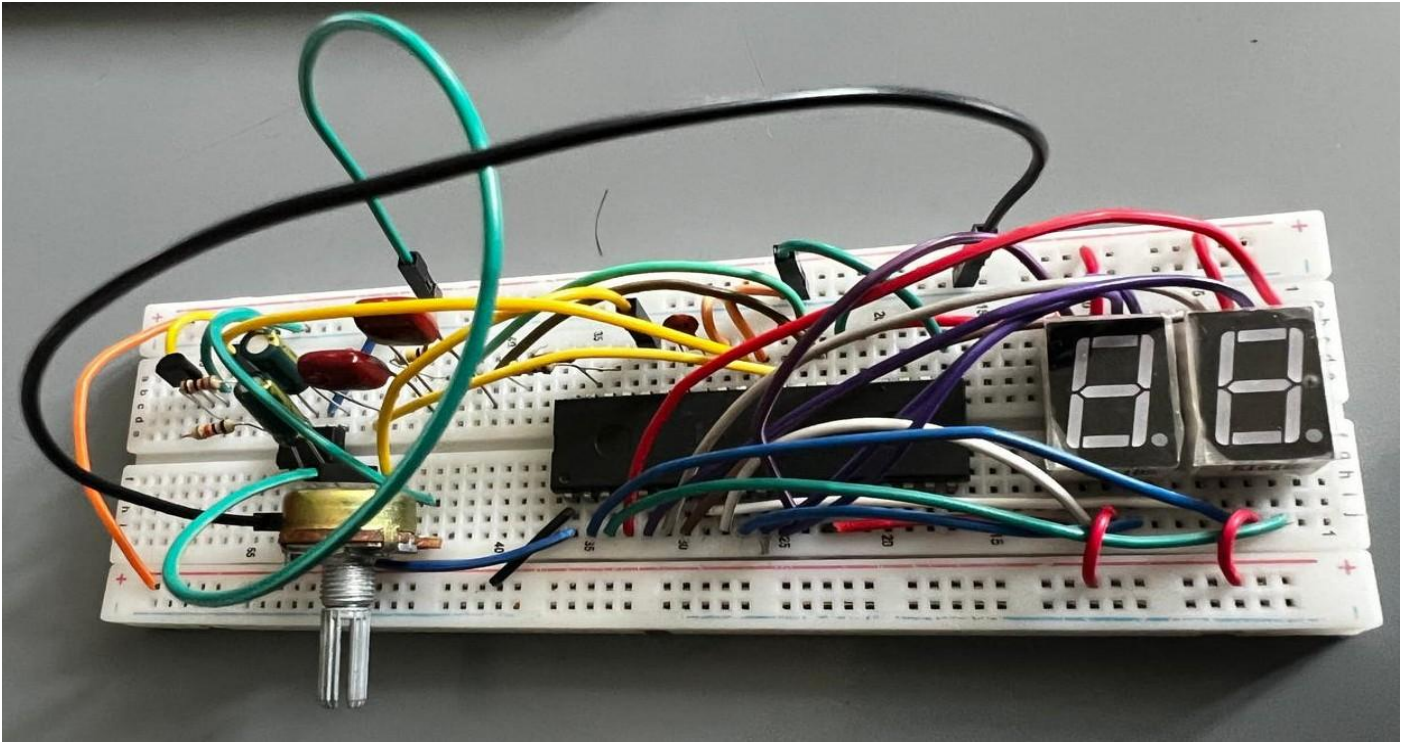


Figure 5: Breadboard implementation of the digital thermometer circuit during hardware testing.

- The practical hardware implementation on a breadboard is shown in Figure 5. Due to the use of long jumper wires and lack of proper grounding, the system exhibited unstable readings.

5.5 Observed Issues During Hardware Testing

The following practical issues were observed:

1. Unstable Display Readings

The 7-segment display values continuously fluctuated by ± 2 to ± 5 counts even at constant temperature.

2. Noise Sensitivity

Touching the breadboard or wires caused the display value to change.

3. Calibration Difficulty

It was difficult to precisely adjust V_{REF} to exactly 100mV due to:

- Potentiometer sensitivity
- Electrical noise
- Breadboard contact resistance

4. Dual Supply Instability

The -5V generated using ICL7660 was not perfectly stable due to:

- Breadboard wiring resistance
- Long jumper wires
- Insufficient decoupling

5.6 Technical Analysis of Instability

The instability occurred mainly due to analog measurement sensitivity.

5.6.1 Breadboard Parasitic Effects

Breadboards introduce:

- Parasitic capacitance ($\sim 2\text{--}20\text{pF}$ between rows)
- Contact resistance
- Loose connections
- Long signal loops

Since LM35 output is very small:

$$10\text{mV} = 1^\circ\text{C}$$

Even a noise of:

$$5\text{mV} \Rightarrow 0.5^\circ\text{C error}$$

This makes the display fluctuate.

5.6.2 Ground Noise Problem

ICL7107 is a precision ADC.

If ground shifts by only:

$$2\text{mV}$$

Then:

$$\text{Error} = \frac{2\text{mV}}{100\text{mV}} \times 100 = 2\%$$

That means:

$$2^\circ\text{C error}$$

Because breadboard does not support proper star grounding, ground noise caused display jitter.

5.6.3 Power Supply Ripple

The ICL7660 charge pump produces switching noise.

Without proper PCB layout and short traces:

$$V_{ripple} \propto \frac{I}{fC}$$

Any ripple directly affects the ADC reference.

5.6.4 Long Signal Wires

The LM35 analog output was connected using long jumper wires.

Long wires act as:

- Antennas
- Noise pickup paths

Induced noise:

$$V_{noise} = L \frac{di}{dt}$$

This caused fluctuating ADC input.

5.7 Why the Circuit Works in Simulation but Not Stable in Hardware

Simulation:

- No parasitic resistance
- No noise
- Ideal ground
- Ideal power supply

Real Hardware:

- Electrical noise
- Contact resistance
- Switching ripple
- Thermal noise

Therefore, the instability is due to practical non-ideal factors, not design error.

5.8 PCB Fabrication Status

Although a professional PCB was fully designed in EasyEDA including:

- Proper star grounding
- Short analog tracks
- Display alignment
- Dual-layer routing
- 3D visualization

The PCB was not fabricated on copper board due to:

1. Time constraints
2. Limited access to PCB etching facilities
3. Difficulty in manually drilling 40-pin DIP footprint
4. Risk of damaging expensive IC components

Therefore, only breadboard implementation was physically tested.

6. DISCUSSION

During the development and hardware implementation of the 2-digit digital thermometer, several practical challenges were encountered. Although the system was fully functional in simulation, hardware testing revealed multiple real-world issues. The challenges and their corresponding engineering solutions are discussed below.

6.1 Challenge 1: Unstable Display Readings

Problem Observed:

- The 7-segment display values fluctuated continuously (± 2 to ± 5 counts) even when the temperature was constant.

Technical Cause:

- The LM35 outputs a very small voltage:

$$10mV = 1^{\circ}C$$

Any small electrical noise (2–5mV) causes:

$$0.2^{\circ}C - 0.5^{\circ}C \text{ error}$$

Additionally:

- Breadboard parasitic capacitance
- Long jumper wires
- ADC high sensitivity
- Lack of proper analog grounding

caused voltage fluctuations at the ADC input.

Engineering Solution:

1. Use a properly fabricated PCB instead of a breadboard.
2. Implement star grounding on PCB.
3. Add a low-pass filter at LM35 output:

$$R = 100\Omega, C = 1\mu F$$

4. Use short analog traces.
5. Place decoupling capacitors close to IC pins.

6.2 Challenge 2: Noise from Dual Power Supply (-5V Generation)

Problem Observed:

- The -5V generated using ICL7660 was not stable and introduced switching noise.

Technical Cause:

- ICL7660 is a charge pump inverter.

Switching ripple voltage:

$$V_{ripple} \propto \frac{I}{fC}$$

On breadboard:

- Long connections
- Poor decoupling
- High impedance ground

caused ripple to affect the ADC reference.

Engineering Solution:

1. Increase charge pump capacitors (e.g., 22 μ F instead of 10 μ F).
2. Add 100nF ceramic capacitors near power pins.
3. Use linear -5V regulator (7905) for better stability.
4. Separate analog and digital grounds.

6.3 Challenge 3: Reference Voltage Calibration Difficulty

Problem Observed:

- It was difficult to adjust the potentiometer to exactly 100mV.
- Small rotation caused large display variation.

Technical Cause:

- The display equation:

$$Display = \frac{V_{IN}}{V_{REF}} \times 100$$

- If VREF error = ± 5 mV:

$$\frac{5}{100} \times 100 = 5\% \text{ error}$$

That means $\pm 5^\circ\text{C}$ error possible.

Potentiometer instability and contact resistance increased variation.

Engineering Solution:

1. Use a multi-turn precision potentiometer.
2. Replace potentiometer with precision voltage reference IC.
3. Add buffering op-amp for reference stability.
4. Use 1% tolerance resistors.

6.4 Challenge 4: Breadboard Parasitic Effects

Problem Observed:

- Touching wires or moving hands near the circuit changed the display value.

Technical Cause:

Breadboards introduce:

- Parasitic capacitance (~10pF)
- Contact resistance
- Inductive loops

Long wires act as antennas and pick up EMI:

$$V_{noise} = L \frac{di}{dt}$$

This directly affects ADC measurement.

Engineering Solution:

1. Use PCB with short, thick traces.
2. Twist power supply wires to reduce EMI.
3. Shield analog input path.
4. Keep LM35 physically close to ADC input.

6.5 Challenge 5: Complex Wiring of DIP-40 IC

Problem Observed:

- The ICL7107 has 40 pins, resulting in complex wiring and possible connection errors.

Technical Cause:

- High pin count increases routing complexity.
- Increased chance of miswiring.
- Large number of display connections.

Engineering Solution:

1. Use PCB to simplify routing.
2. Label wires carefully during prototyping.
3. Use ribbon cable for display connections.
4. Test each segment individually before full integration.

6.6 Challenge 6: Ground Loop Noise

Problem Observed:

- Display jitter increased when digital segments were switching.

Technical Cause:

- When display segments turn ON, current spikes occur.

If ground impedance is not low:

$$V_{error} = I \times R$$

Even small ground resistance creates measurable error at ADC.

Engineering Solution:

1. Implement single-point star grounding.
2. Use thicker ground traces.
3. Separate analog and digital ground paths.
4. Use ground plane in PCB design.

6.7 Challenge 7: Time and Fabrication Constraints

Problem Observed:

- The PCB could not be fabricated on copper board.

Reason:

- Limited time before submission
- No PCB etching facility access
- Difficulty drilling 40-pin DIP accurately
- Risk of damaging components

Proposed Solution:

If given additional time:

1. Fabricate PCB using professional service.
2. Assemble components carefully.
3. Perform final calibration.
4. Enclose in protective casing.

6.8 Lessons Learned

Through this project, the following engineering lessons were learned:

1. Precision analog circuits should not be tested on breadboards.
2. Proper PCB layout is critical for ADC circuits.
3. Grounding design is as important as circuit design.
4. Simulation does not represent real-world noise.
5. Reference voltage stability determines measurement accuracy.

6.9 Improvements for Future Work

If further development is allowed, the following improvements will be implemented:

1. Use Dedicated Voltage Reference IC

Example:

- LM4040-2.5V precision reference with a resistor divider to generate accurate 100mV.
- Instead of potentiometer

2. Use Linear Negative Regulator

Instead of ICL7660, use:

- 7905 regulator
- With transformer supply

3. Fabricate PCB

Advantages:

- Reduced noise
- Short analog traces
- Proper star ground
- Stable readings

4. Add Filtering Capacitor at LM35 Output

Add:

$$C = 1\mu F$$

To reduce noise at sensor output.

5. Use Shielded Wire for Sensor

Prevents EMI pickup.

7. CONCLUSION

This project successfully designed and analyzed a 2-digit digital thermometer using the LM35 temperature sensor and the ICL7107 analog-to-digital converter with integrated 7-segment display drivers. The system was mathematically modeled to ensure direct temperature-to-display scaling, where a reference voltage of 100 mV was carefully calculated so that 10 mV from the LM35 corresponds to 1°C on the display. The complete circuit was verified through Proteus simulation, confirming correct voltage scaling and accurate temperature representation.

A professional PCB layout was developed in EasyEDA following proper analog design principles, including star grounding, short analog signal routing, decoupling capacitor placement, and horizontal alignment of the displays. Although the practical implementation on a breadboard exhibited unstable readings due to noise, parasitic effects, and power supply ripple, the analysis demonstrated that these issues were related to real-world non-idealities rather than design errors.

Overall, this project enhanced understanding of precision analog measurement systems, ADC operation, reference voltage scaling, grounding techniques, and PCB design considerations. The work successfully met the academic objectives while highlighting the importance of proper hardware layout in achieving stable and accurate measurements.

8. REFERENCES

Texas Instruments. LM35 Precision Centigrade Temperature Sensors Datasheet.

Renesas Electronics. ICL7107 3½ Digit A/D Converter with LED Display Driver Datasheet.

Renesas Electronics. ICL7660 CMOS Voltage Converter Datasheet.

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